AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Project Domain

[Excel / PowerBIwithCopilot]

A Project Report

submitted in partial fulfilmentoftherequirements of the AICW Project

by:

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EMPLOYEE PAYROLL DATA CLEANING AND FINANCIAL ANALYSIS

PROBLEM STATEMENT

Organizations rely heavily on payroll systems to manage employee salaries, bonuses, and tax deductions. However, payroll data is often affected by inconsistencies such as missing values, duplicate records, incorrect data formats, and calculation errors. These issues can lead to inaccurate salary processing, financial mismanagement, and poor decision-making.

In many organizations, payroll datasets are large and complex, making manual verification difficult and time-consuming. Errors such as incorrect tax deductions or duplicate employee records may go unnoticed, leading to financial discrepancies.

Additionally, while Artificial Intelligence tools are becoming widely available, many users lack the skills to utilize them effectively for data analysis and reporting. This creates a gap between data availability and actionable insights.

This project aims to address these challenges by performing data cleaning and financial analysis on an employee payroll dataset. By applying Excel-based techniques and AI-assisted insights, the project demonstrates how structured data processing can improve accuracy, efficiency, and decision-making in payroll management.

OBJECTIVES

- To collect and explore a payroll dataset containing employee salary details.
- To identify and correct errors such as missing values, duplicates, and incorrect data formats.
- To perform financial analysis on employee salaries using Excel formulas and pivot tables.
- To create visual representations such as charts and dashboards for better understanding.
- To use an AI chatbot to generate insights and summaries from the dataset.
- To improve data accuracy and demonstrate efficient payroll data handling techniques.

METHODOLOGY / IMPLEMENTATION PLAN

#	Phase	Description
1	**Data Collection & Exploration**	Obtain the raw dataset (CSV / Excel) from Kaggle, Edunet, or any suitable source. Understand field names, data types, and row count. For AI prompt projects, define the use case and target audience instead.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle null or missing values, fix incorrect data types, and standardize data and text formats using Excel (Remove Duplicates, IF, IFERROR, Text-to-Columns) or Power Query. For prompt-based projects, draft and structure initial prompts using Role, Context, Task, Format, and Tone components.
3	**Analysis & AI Integration**	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP), or Power BI DAX measures to derive key insights. Where applicable, use Microsoft Copilot or an AI Chatbot to generate narratives, summaries, or content outputs from the data or prompts.
4	**Visualization / Output**	Build interactive dashboards with charts, KPI cards, and slicers in Power BI or Excel. For AI-assisted projects, compile generated content outputs such as reports, marketing copy, or summaries. Document all AI prompts used alongside their outputs.
5	**Documentation & Submission**	Write the project report, upload all files to a GitHub repository, and record a video demonstration. Submit the GitHub link, video link, and dashboard link (if applicable) for evaluation.

- This project follows a systematic approach to clean and analyze employee payroll data using Microsoft Excel. Initially, a raw dataset containing errors such as missing values, duplicate records, and incorrect data formats was examined. Data cleaning techniques were applied to identify and correct these issues, including handling missing values, removing duplicates, and standardizing data types.

- After cleaning, formulas were used to calculate salary components, and pivot tables were created for analysis. Visualizations such as charts were used to represent the data clearly. Each step of the process was supported with screenshots to demonstrate the transformation from raw data to a structured and accurate dataset.

Phase 1: Data Collection & Exploration

- A payroll dataset was created to simulate real-world employee salary records. The dataset includes fields such as Employee ID, Name, Department, Basic Salary, Bonus, Tax, Net Salary, and Payment Date.
- Initial exploration revealed several inconsistencies such as missing values, incorrect data entries, and duplicate records.

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID	Employee_Name	Department	Designation	Date_of_Joining	Basic_Salary	Allowances	Bonus	Tax_Deduction	Net_Salary	Payment_Date
2	EMP001	Rahul Sharma	HR	HR Executive	15-01-2020	45000	5000	4000	4500	49,500	31-05-2024
3	EMP002	Priya Verma	Finance	Accountant	20/02/2019	52,000	6000	5000	5200	57,800	31/05/2024
4	EMP002	Priya Verma	Finance	Accountant	20-02-2019	52000	6000	5000	5200	57800	31-05-2024
5	EMP003	Amit Kumar	IT	Software Engineer	10-03-21	60,000	700 0	6000	6000	67000	31-05-2024
6	EMP004	Anjali Singh	Marketing	Marketing Executive	05-07-2020	47000	5500	4000	4700	50,800	31/05/2024
7	EMP005	Vikram Patel	Sales	Sales Executive	12/11/2018	48,000.00	5000	4500	4800	31,900.00	31-05-2024
8	EMP006	Neha Gupta	HR	HR Executive	18-06-2019	46000	5000	4000	4600	50400	31/05/2024
9	EMP007	Saurabh Yadav	IT	System Analyst	22-08-2021	62000	7500	6000	6200	69,300	31-05-2024
10	EMP008	Pooja Mehta	Finance	Finance Executive	30/09/2018	51000	6000	5000	5100	57900	31-05-2024
11	EMP009	Rohit Das	IT	Software Engineer	14-04-2020	58,000	7000	5500	5800	64,700	31/05/2024
12	EMP010	Kavita Joshi	Marketing	Marketing Executive	25/01/2019	48000	5,500	4500	4800	52400	31-05-2024
13	EMP011	Deepak Sharma	Sales	Sales Executive	11-05-2021	49000	5000	4500	4900	52,600	31-05-2024
14	EMP012	Sneha Reddy	HR	HR Executive	19/12/2018	45,000	5000	4000	4500	49500	31/05/2024
15	EMP013	Arjun Nair	IT	Software Engineer	07-03-2020	61000	7500	6000	6100	68,400	31-05-2024
15	EMP014	Meera Iyer	Finance	Accountant	29-07-2019	50,000	6000	5000	5000	57,000	31-05-2024
17	EMP015	Manish Kumar	Sales	Sales Executive	03/09/2020	47000	5000	4500	4700	51800	31/05/2024
18	EMP016	Riya Malhotra	Marketing	Marketing Executive	21-02-2021	49,000	5500	4500	4900	53200	31-05-2024
18	EMP017	Mohit Verma	IT	System Analyst	16-10-2019	60000	7000	6000	6000	67000	31-05-2024
20	EMP018	Divya Jain	Finance	Finance Executive	04/06/2020	52000	6000	5000	5200	57800	31/05/2024
21	EMP019	Kunal Singh	Sales	Sales Executive	28-11-2018	48,000	5000	4500	4800	51,900	31-05-2024
22	EMP020	Simran Kaur	HR	HR Executive	09-01-2021	46000	5000	4000	4600	50400	

Figure 1: Data Collection (Uncleaned)

Phase 2: Data Cleaning & Preprocessing

The dataset was cleaned using various Excel tools and techniques to ensure accuracy and consistency.

1. Identifying Errors

Errors such as missing salary values, duplicate records, invalid text entries, and incorrect date formats were identified.

2. Removing Duplicates

Duplicate employee records were removed using Excel’s “Remove Duplicates” feature.

3. Handling Missing Values
Missing salary values were filled using appropriate estimation methods.

4. Correcting Data Types
Text values in numeric columns were converted into a proper numeric format.

5. Fixing Invalid Values
Negative and unrealistic values such as incorrect tax entries were corrected.

6. Fixing Date Format
Incorrect and inconsistent dates were corrected.

7. Final Clean Dataset
After all corrections, the dataset became structured and ready for analysis.

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID	Employee_Name	Department	Designation	Date_of_Joining	Basic_Salary	Allowances	Bonus	Tax_Deduction	Net_Salary	Payment_Date
2	EMP001	Rahul Sharma	HR	HR Executive	15-01-2020	45,000.00	5,000.00	4,000.00	4,500.00	49,500.00	31-05-2024
3	EMP002	Priya Verma	Finance	Accountant	20-02-2019	52,000.00	6,000.00	5,000.00	5,200.00	57,800.00	31-05-2024
4	EMP003	Amit Kumar	IT	Software Engineer	10-03-2021	60,000.00	7,000.00	6,000.00	6,000.00	67,000.00	31-05-2024
5	EMP004	Anjali Singh	Marketing	Marketing Executive	05-07-2020	47,000.00	5,500.00	4,000.00	4,700.00	50,800.00	31-05-2024
6	EMP005	Vikram Patel	Sales	Sales Executive	12-11-2018	48,000.00	5,000.00	4,500.00	4,800.00	31,900.00	31-05-2024
7	EMP006	Neha Gupta	HR	HR Executive	18-06-2019	46,000.00	5,000.00	4,000.00	4,600.00	50,400.00	31-05-2024
8	EMP007	Saurabh Yadav	IT	System Analyst	22-08-2021	62,000.00	7,500.00	6,000.00	6,200.00	69,300.00	31-05-2024
9	EMP008	Pooja Mehta	Finance	Finance Executive	30-09-2018	51,000.00	6,000.00	5,000.00	5,100.00	57,900.00	31-05-2024
10	EMP009	Rohit Das	IT	Software Engineer	14-04-2020	58,000.00	7,000.00	5,500.00	5,800.00	64,700.00	31-05-2024
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21	EMP020	Simran Kaur	HR	HR Executive	09-01-2021	46,000.00	5,000.00	4,000.00	4,600.00	50,400.00	31-05-2024

Figure 2: Cleaned Dataset Version

Phase 3: Analysis & Ai Integration

After cleaning, financial analysis was performed to extract meaningful insights. Excel functions such as SUM, AVERAGE, and IF were used to calculate salary-related metrics. Pivot tables were created to analyze department-wise salary distribution. for assisting with these tasks Ai integration helps. Example ; ChatGPT.

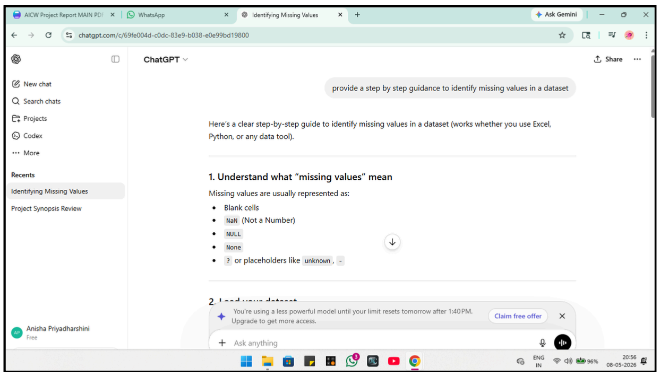

Figure 3: Ai Integration (ChatGPT)

Phase 4: Visualization / Output

Charts and graphs were created to visually represent payroll insights.

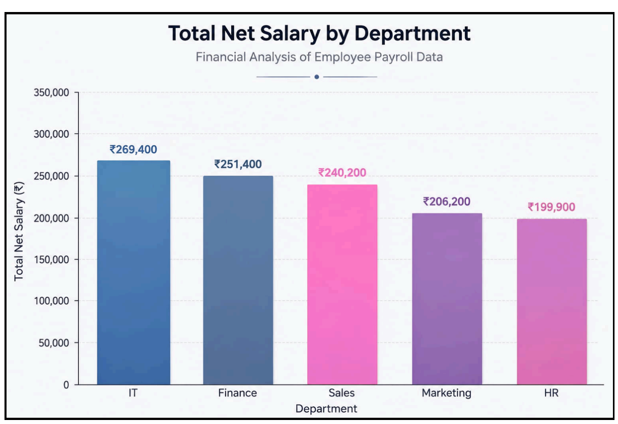

Figure 4: Bar Chart

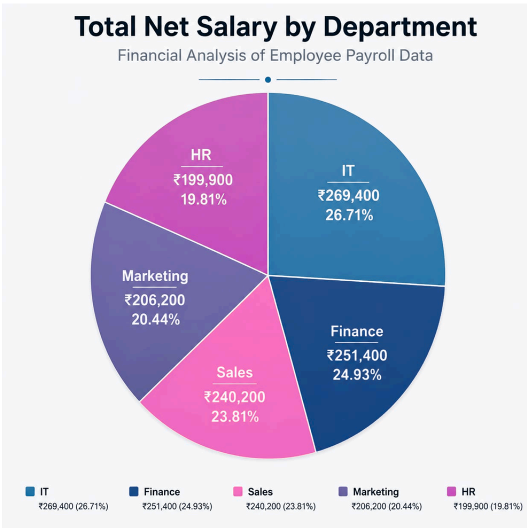

Figure 5: Pie Chart

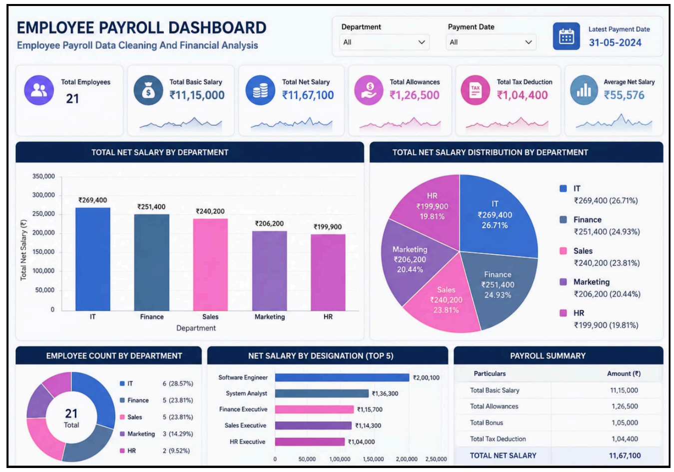

Figure 6: Dashboard

Phase 5: Documentation & Submission

- All steps of the project were documented, and outputs were prepared for submission including dataset, analysis, and report.
- The cleaned dataset and analysis results were verified to ensure accuracy and consistency before submission.

RESOURCES AND TIMELINE

A. Hardware & Software Requirements

- Laptop with minimum 4 GB RAM
- Microsoft Excel (2019 or later)
- AI Chatbot (ChatGPT)
- Internet connection
- Web browser

B. Dataset

- The dataset used in this project is a simulated payroll dataset created to represent real-world employee salary records. It includes details such as employee information, salary components, tax deductions, and payment dates.
- The dataset was intentionally designed with errors such as missing values, duplicates, and incorrect formats to demonstrate the data cleaning process.

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

EXPECTED OUTCOMES / CONCLUSION

The project is expected to result in a clean and structured payroll dataset with all inconsistencies removed. Financial analysis will provide insights into salary distribution, department-wise expenses, and employee compensation patterns.

The use of AI tools will enhance the ability to generate insights and summaries efficiently. The final output will include a well-documented report, visual charts, and a clear demonstration of data cleaning and analysis techniques.

This project highlights the importance of data accuracy and the role of data analytics in improving organizational decision-making.

REFERENCES

- Microsoft Excel Documentation
- Kaggle Dataset Resources
- Microsoft Copilot & ChatGPT (AI Chatbot)
- Online tutorials and learning materials

LINKS

- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com

- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets

- Open AI, ChatGPT. Available at: https://chatgpt.com/

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org

- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning

- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

- Project Github Available at : https://github.com/anishapriyadharshini/Employee-Payroll-Data-Cleaning-And-Financial-Analysis-.git

Thank You